JACKSON CMELAK

Software Engineer

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EDUCATION

B.S. Computer Science **Sept 2022 – Dec 2025**
University of California, Davis *Davis, CA*

- Relevant coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Computer Architecture, Database Systems, Scientific Computation

EXPERIENCE

Freelance Web Developer **Dec 2025 – Present**
Self-Employed *Laguna Niguel, CA*

- Own the full lifecycle of production business websites in Svelte and TypeScript, from requirements gathering through design, implementation, testing, deployment, and ongoing maintenance.
- Run backend data storage and cloud hosting on Zoho Catalyst, integrating RESTful APIs and third-party SDKs, and troubleshoot and resolve production issues for live client sites.

Data Analyst Intern **Jun 2022 – Aug 2022**
US Cryotherapy *San Ramon, CA*

- Consolidated five years of operational data from point-of-sale, marketing, and web analytics systems into one dataset, then built regression models in Python (NumPy, Pandas) to drive business recommendations.
- Presented findings in a full-day workshop to ownership and management, who built the following year's advertising strategy around them.

Robotics Programmer **Nov 2015 – Apr 2017**
FIRST Robotics Team 1458, Red Tie Robotics *Danville, CA*

- Wrote movement and computer vision code in Java and OpenCV for a competition robot across two seasons, delivering working software on hard deadlines with physical hardware on the other end.

PROJECTS

UDP Telemetry Receiver **C++17 / CMake / BSD Sockets**
Personal Project *Completed*

- Designed a fixed 28-byte binary protocol for simulated GPS/IMU sensor data and wrote a C++17 sender and receiver that exchange it over UDP sockets.
- Implemented table-driven CRC-32 and explicit network byte ordering by hand; the parser returns std::optional and rejects any datagram that fails size, magic-number, or checksum validation.
- Factored the protocol codec into a CMake static library shared by both endpoints so the wire format cannot drift.

Bone Fracture Detector **Python / PyTorch / scikit-learn**
Computer Vision Project *Completed*

- Fine-tuned a pre-trained ResNet via transfer learning to detect fractures in X-ray images, reaching 83.4% accuracy (ROC AUC 0.834) on a 338-image test set.
- Built a hand-engineered-feature SVM baseline first (58.6% accuracy) to measure the 25-point gain, and benchmarked results against a published baseline from the imaging literature.

BoosterTutor **Python / FastAPI / PostgreSQL / SvelteKit**
AI-Powered Game Analysis App *In Development*

- Building a full-stack coaching app (FastAPI, SQLAlchemy, PostgreSQL, SvelteKit) in which Python computes every verdict from real win-rate data and the LLM only explains it, keeping model output grounded and free of invented numbers.
- Grades each pick against ten competing strategy hypotheses for the user's pool; tested with pytest, mypy, Vitest, and Playwright, with design changes tracked in architecture decision records.

Garbage Classifier **Python / TensorFlow / Streamlit**
UC Davis Machine Learning Team Project *Completed*

- Compared an SVM, random forest, and CNN for six-class image classification of waste materials; the CNN reached 74% accuracy, 18 points above the best classical model, and was deployed in a Streamlit demo app.

TECHNICAL SKILLS

Languages: C++, C, Python, Java, TypeScript, SQL

Frameworks & Tools: Linux, Git, CMake, PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas, FastAPI, SQLAlchemy, PostgreSQL, SQLite, Svelte/SvelteKit, Node.js, pytest, Playwright

Practices: Systems Programming, Network Protocols, Computer Vision, Machine Learning, RESTful API Design, Unit Testing, ETL / Data Pipelines, LLM API Integration, AI-Assisted Development